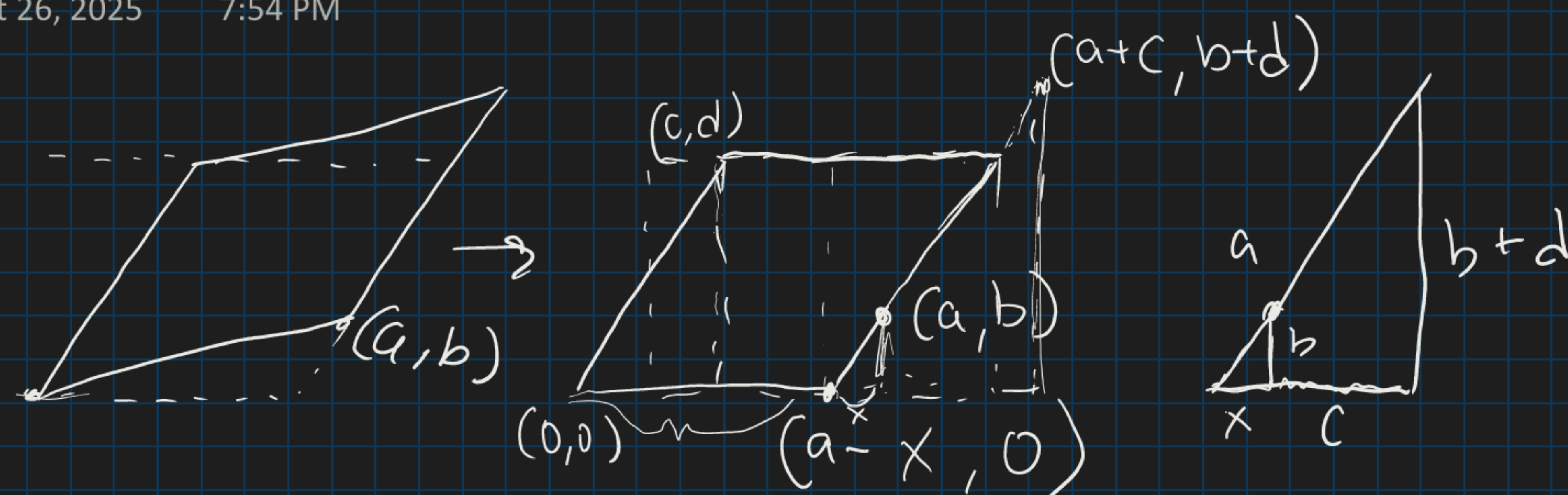


1. a)



$$hw$$

$$h=d$$

$$w=a-x$$

by similar  $\triangle$ :  $\frac{x}{x+c} = \frac{b}{b+d}$

$$x(b+d) = b(x+c)$$

$$xb + xd = bx + bc$$

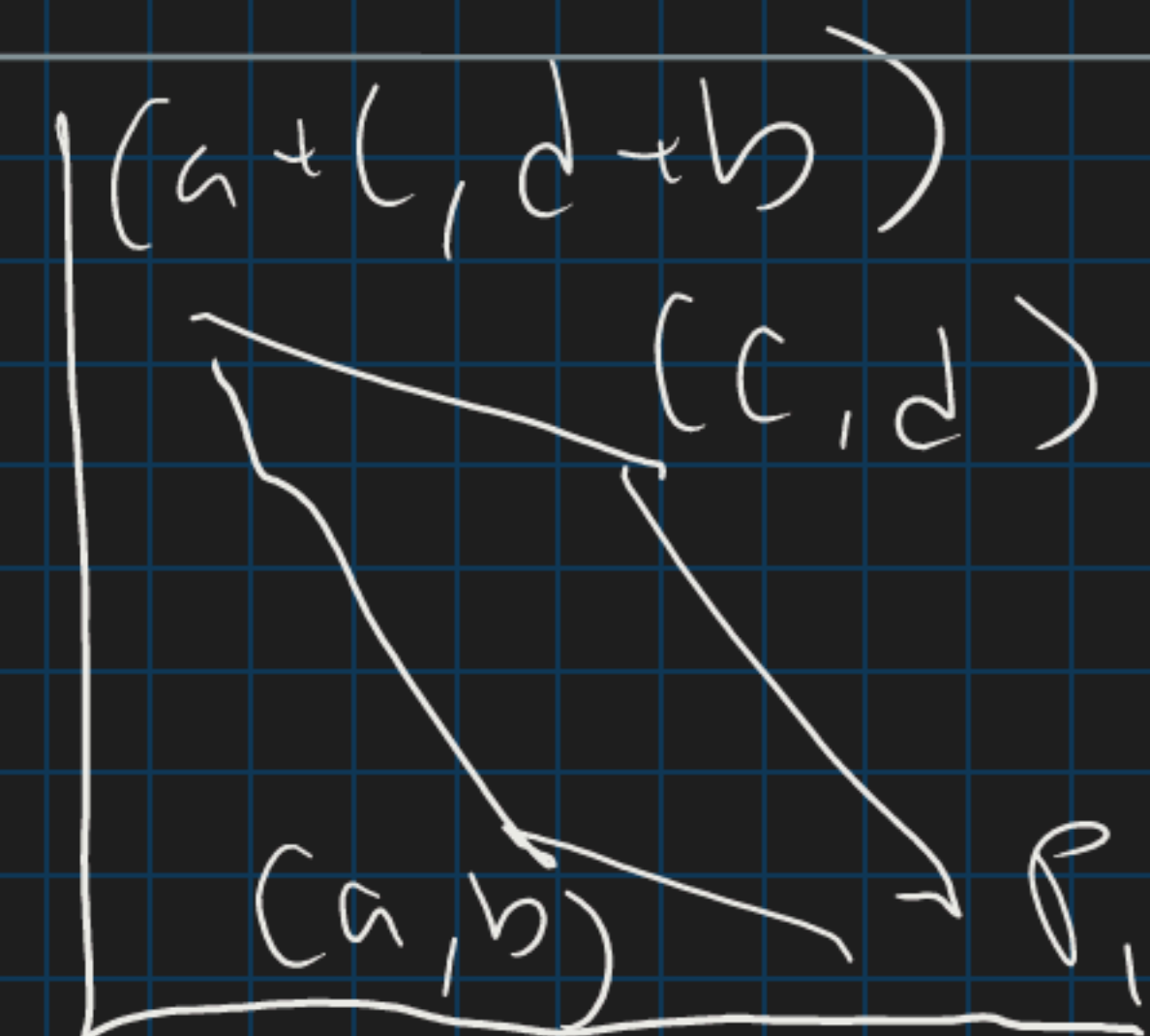
$$x = \frac{bc}{d}$$

$$h=d$$

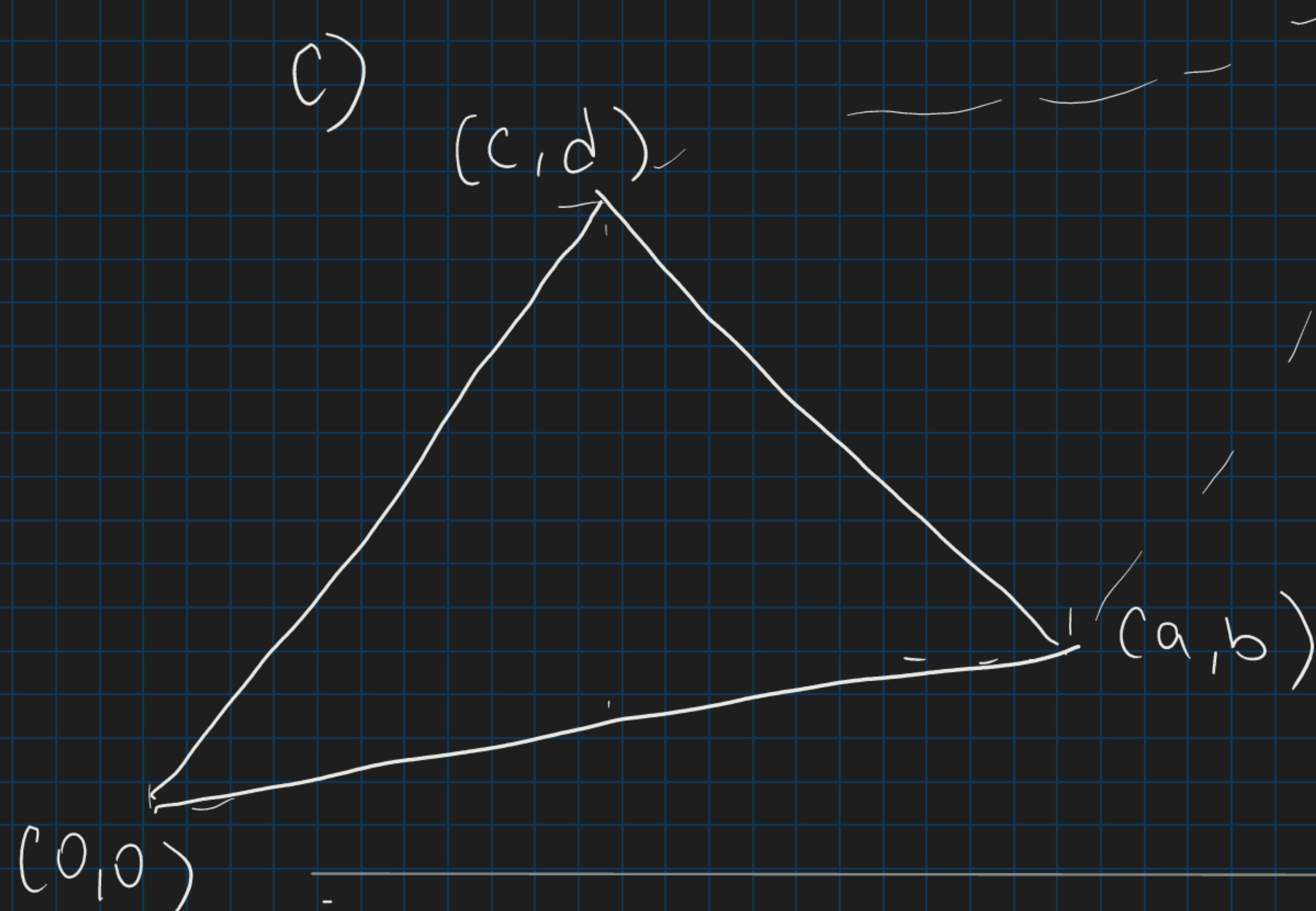
$$h \cdot w = d \cdot \left(a - \frac{bc}{d}\right)$$

$$hw = ad - bc = P_{\text{area}}$$

b) if  $bc > ad$  then  $ad - bc$  would be negative, and area can't be negative. If you reflect the original parallelogram over  $y=0$ , (then translate the parallelogram so no values are negative) then you have a parallelogram with the same area but with  $bc > ad$ . the sign is thus negative, so the absolute value would correct this so the area is positive and actually works physically

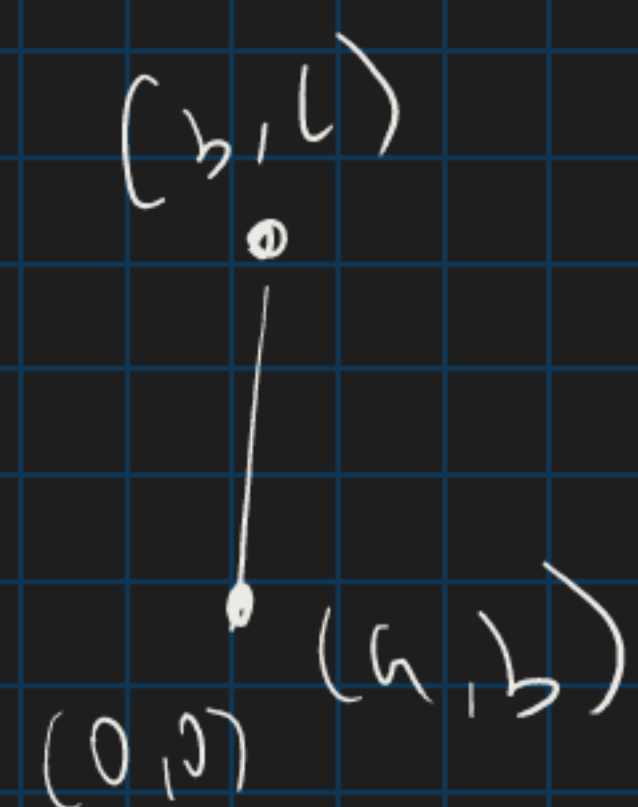


c)



d) If you reflect the triangle so that there are 2 over the line  $(c,d)$  to  $(a,b)$ , then the resulting combined shape is a parallelogram. The equation to compute area of a parallelogram is  $|ad - bc|$ , and since this shape is made of 2 congruent triangles,  $\frac{1}{2}|ad - bc|$  is the area of one triangle.

d)



The figure  $P$  is a line